Do not write outside the box

Please write clearly in block capitals.

Centre number | | | | | |

Candidate number | | | | |

Surname

Forename(s)

Candidate signature

Combined Science: Trilogy

Biology Paper 1 topics (4.1 to 4.4): practice paper 2

Foundation Tier

Suggested time: about 30 minutes. There is no timer.

For Examiner's Use	
Question	**Mark**
1	
2	
3	
4	
TOTAL	

Materials

For this paper you must have:

- a ruler
- a scientific calculator

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer all questions in the spaces provided.
- Do all rough work in this book. Cross through any work you do not want to be marked.
- In all calculations, show clearly how you work out your answer.

Information

- The maximum mark for this paper is 30.
- The marks for questions are shown in brackets.
- You are expected to use good English and clear presentation in your answers.

This is a practice paper written for Dougie in the style of the GCSE exam. It is not an exam board paper, and it is not a paper from any past exam series.

0 1 . 1 Which part of a cell controls its activities? Tick (✓) one box.

Cell membrane ☐

Cytoplasm ☐

Nucleus ☐

Ribosome ☐

[1 mark]

0 1 . 2 Bacterial cells are prokaryotic. Give one way a bacterial cell is different from an animal cell.

[1 mark]

0 1 . 3 A cell is 0.02 mm wide. An image of the cell is 4 mm wide. Calculate the magnification. Use: magnification = size of image ÷ size of real object. Show your working.

Magnification = × .....

[2 marks]

0 1 . 4 Describe what osmosis is.

[2 marks]

0 1 . 5 A plant cell is put into a very concentrated salt solution. What happens to the cell? Tick (✓) one box.

It bursts. ☐

It gains water and swells. ☐

It loses water and shrinks. ☐

Nothing happens. ☐

[1 mark]

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0 2 . 1 Which enzyme breaks down protein? Tick (✓) one box.

Amylase ☐

Bile ☐

Lipase ☐

Protease ☐

[1 mark]

0 2 . 2 Name the products of protein digestion.

[1 mark]

0 2 A student tested a food. The table shows the results.

Test	Colour at the end
Iodine solution	Blue-black
Benedict's solution (heated)	Blue
Biuret reagent	Purple

0 2 . 3 Which two types of food molecule does the food contain? Use the results in the table.

[2 marks]

0 2 . 4 Explain why amylase can break down starch but cannot break down protein.

[2 marks]

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0 2 . 5 A student wants to find out how temperature affects how quickly amylase breaks down starch. Describe a method the student could use. Include how they would make it a fair test.

[6 marks]

0 3 . 1 Salmonella is a pathogen that causes food poisoning. What type of pathogen is it? Tick (✓) one box.

- Bacterium ☐
- Fungus ☐
- Protist ☐
- Virus ☐

[1 mark]

0 3 . 2 Give two ways the body stops pathogens getting in.

[2 marks]

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0 3 . 3 A doctor gives a patient antibiotics for a bacterial infection. Explain why the doctor would not give antibiotics to someone with a cold.

[2 marks]

0 3 . 4 What do painkillers do? Tick (✓) one box.

They cure the disease. ☐

They kill bacteria. ☐

They kill viruses. ☐

They relieve the symptoms. ☐

[1 mark]

0 4 . 1 Complete the word equation for photosynthesis: carbon dioxide + water → _____ + oxygen

[1 mark]

0 4 . 2 Give two factors that affect the rate of photosynthesis.

[2 marks]

0 4 . 3 During hard exercise, muscles can respire anaerobically. Name the substance made by anaerobic respiration in muscles.

[1 mark]

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0 4 . 4 Give one reason why aerobic respiration is better for the body than anaerobic respiration.

[1 mark]

END OF QUESTIONS

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